



# LaLonde, Dehejia– Wahba, and Why the $2 \times 2$ Still Matters

*Selection on observables, in the case that  
built the literature*

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University of Glasgow · May 2026

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2026

LESSON

# 1. LaLonde 1986

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# The National Supported Work Demonstration

A federally funded **randomized** job-training program.

1975–1979. Four target groups, including  
disadvantaged adult men.

*Random assignment to treatment vs. control. Outcome: real earnings in 1978.*

## LaLonde's experimental benchmark

$$\widehat{ATT}^{\text{RCT}} \approx \$886$$

1978 dollars. Adult-male subsample.

*Random assignment  $\Rightarrow$  no selection. This is the truth.*

## Then LaLonde threw away the controls

He kept the treated — the 297 NSW men.

He replaced the controls with samples from the  
**CPS** and the **PSID**.

*A simulation of what an applied researcher would face —  
no experiment, just observational data and methods.*

## Then he tried every method going at the time

- OLS with controls
- Two-step Heckman selection corrections
- Fixed effects on pre-program earnings
- Difference-in-differences

*Could any of them recover \$886?*

No

Estimates ranged from  $-\$16,000$  to  $+\$3,000$ .

Most were negative. Most missed the benchmark by orders of magnitude.

*LaLonde (1986), AER 76(4): 604–620.*

## The headline that shook applied econometrics

*“The econometric procedures  
cannot replicate the experimental results.”*

*This is the paper that made program evaluation  
move toward randomization and design.*



LESSON

# 2. Dehejia & Wahba 2002

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## Dehejia & Wahba reopened the case

**Rajeev Dehejia** and **Sadek Wahba**,  
*Review of Economics and Statistics* (2002).

They asked: does **propensity score matching**  
do better than LaLonde's methods?

## First, they re-cut the experimental sample

They restricted to NSW men with earnings observed in  
**both 1974 and 1975.**

A smaller, slightly different population.

*This is the subsample the homework uses: `lalonge_exp_panel.dta`.*

## A new experimental benchmark

$$\widehat{ATT}_{D\&W}^{RCT} \approx \$1,794$$

Roughly **\$1,700** in the way we usually report it.

*Why bigger than LaLonde's \$886? Different subsample, different ATT.*

## Why the headlines differ

- Both numbers come from the *same* NSW experiment.
- LaLonde used the full adult-male sample.
- D&W used a *subset* — only men with two years of pre-program earnings.
- Subsetting on a covariate changes the population whose ATT you're estimating.
- Same identification (random assignment). Different target group.

*This is not a contradiction — it is the ATT defined on different units.*

## D&W's quasi-experimental result

With **propensity-score matching** on  
CPS / PSID controls...

they recovered  $\approx$  \$1,700.

*A constructive answer to LaLonde: design plus matching can work.*

LESSON

# 3. Why the ATT didn't move

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## A puzzle

We used **non-experimental controls** (CPS, PSID)  
in the larger LaLonde–D&W literature.

And yet the ATT we target is still  $\approx \$1,700$ .

*Why doesn't the ATT change when we change the controls?*



Because the ATT is a property of the treated

$$ATT = \mathbb{E}[Y^1 - Y^0 \mid D = 1, \text{Post}]$$

Only the  $D = 1$  side appears.

The control units are just a *tool* to learn about  $\mathbb{E}[Y^0 \mid D=1, \text{Post}]$ .

## Random assignment gave us a clean tool

Under random assignment:

$$\mathbb{E}[Y^0 \mid D=1] = \mathbb{E}[Y^0 \mid D=0]$$

So  $\mathbb{E}[Y \mid D=0, \text{Post}]$  *is* the counterfactual we need.

No selection bias.

## Non-experimental controls break that equality

CPS men and NSW participants are not interchangeable.

$$\mathbb{E}[Y^0 \mid D=1] \neq \mathbb{E}[Y^0 \mid D=0]$$

Selection bias enters.

*NSW men had worse pre-program earnings. The CPS pool didn't.*

## But the *target* hasn't changed

$\mathbb{E}[Y^0 \mid D=1, \text{Post}]$  is a feature of the NSW men.

It does not depend on who we picked as controls.

What the controls control is *our estimate* of that target —  
and the size of the bias.

## So what is DiD for, in this story?

- In the RCT sample: DiD just *confirms* the post-period difference-in-means.
- In the non-experimental sample: DiD tries to *net out* the pre-existing level gap between NSW and CPS men.
- The truth is fixed at \$1,700. The question is whether DiD's parallel-trends assumption is enough to recover it.

*Did it?*

## The benchmark and the challenge

RCT benchmark: \$1,794

Naive DiD on the non-experimental panel: \$3,621

We can do better than additive controls. Let's go to the code.

LESSON

# 4. Now make it harder

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## Throw away the experimental controls

Keep the 297 NSW treated men.

Replace the controls with a random sample of  
American households from the **CPS**.

*Data: lalonde\_nonexp\_panel.dta. Same treated rows. New controls.*



## Selection is severe

|                              | 1974      | 1975      | 1978     |
|------------------------------|-----------|-----------|----------|
| NSW (treated, $n = 185$ )    | \$2,096   | \$1,532   | \$6,349  |
| CPS (control, $n = 15,992$ ) | \$14,017  | \$13,651  | \$14,847 |
| Diff                         | -\$11,921 | -\$12,119 | -\$8,498 |

$$\mathbb{E}[Y^0 \mid D=1] \ll \mathbb{E}[Y^0 \mid D=0]$$

*The level gap is huge — and so is the growth-rate gap. DiD doesn't fix that.*

## The 2×2: where \$3,621 comes from

|                              | 1974     | 1975 (pre) | 1978 (post) |
|------------------------------|----------|------------|-------------|
| NSW (treated, $n = 185$ )    | \$2,096  | \$1,532    | \$6,349     |
| CPS (control, $n = 15,992$ ) | \$14,017 | \$13,651   | \$14,847    |

**NSW change:**  $\$6,349 - \$1,532 = \$4,817$

**CPS change:**  $\$14,847 - \$13,651 = \$1,196$

**DiD**  $\$4,817 - \$1,196 = \$3,621$

$\$3,621 \neq \$1,794$

*NSW men bounce back fast from a deep dip. CPS men don't. DiD attributes the entire gap in growth rates to the program.*

## Naive DiD on this sample

$$\hat{\delta}_{\text{naive DiD}} = +\$3,621$$

*Right sign. Wrong magnitude. The benchmark is \$1,794.*

*NSW men had unusually low pre-earnings (\$1,532) and bounced back fast.  
CPS controls didn't. DiD attributes the gap in growth rates entirely to treatment.*

LESSON

# 5. Run the specs

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## Spec 0: Unconditional parallel trends, no covariates

**Assumption.** Naive parallel trends:  $E[\Delta Y(0) \mid D=1] = E[\Delta Y(0) \mid D=0]$  unconditionally.

**Run:**

```
reg re i.post##i.ever_treated, robust
```

**Compare to \$1,794.**

## Spec A: Conditional PT, additive $X$

**Assumption.** Conditional parallel trends with one slope per covariate (pre = post).

**Run:**

```
global X "age_agesq_educ_educsq_marr_nodgree_black_hisp_re74_re75_u  
reg re i.post##i.ever_treated $X, robust
```

**Compare to \$1,794.**

## Spec B: Conditional PT, Post $\times X$

**Assumption.** Conditional parallel trends, allowing  $X$ -effects to differ pre vs post. Use ## so the spec includes *both* the pre-period  $X$  effect and its post-period change.

**Run:**

```
reg re i.post##i.ever_treated ///  
    i.post##c.age i.post##c.agesq i.post##c.educ i.post##c.educsq ///  
    i.post##i.marr i.post##i.nodegree i.post##i.black i.post##i.hisp ///  
    i.post##c.re74 i.post##c.re75 i.post##i.u74 i.post##i.u75, robust
```

**Compare to \$1,794.** *Single-# forces pre-period  $X$  effect to zero — not what we want.*

## Spec C: Saturated TWFE / OR (HIT 1997)

**Assumption.** Conditional parallel trends; impute  $Y(0)$ 's trend on controls only and average over treated.

**Run — by hand:**

```
reshape wide re, i(id) j(post)
gen dy = re1 - re0

reg dy $X if ever_treated == 0
predict dy_hat
gen delta_hat = dy - dy_hat if ever_treated == 1
summarize delta_hat if ever_treated == 1
```

**Cross-check with drdid:**

```
drdid re $X, time(year) ivar(id) tr(ever_treated) reg
```

**Compare to \$1,794.** *Two numbers should agree to the dollar.*



## Spec D: IPW (Abadie 2005)

**Assumption.** Conditional PT; correct propensity score. Treated weighted by 1; controls weighted by  $\hat{p}(X)/(1 - \hat{p}(X))$ .

### Run — by hand:

```
logit ever_treated $X
predict phat, pr
quietly sum ever_treated
local pr_treat = r(mean)
gen w_ipw = (ever_treated - phat) / (1 - phat) / 'pr_treat'
gen ipw_dy = w_ipw * dy
summarize ipw_dy
```

### Cross-check with drdid:

```
drdid re $X, time(year) ivar(id) tr(ever_treated) ipw
```

### Compare to \$1,794.

## Spec E: DR-DiD (Sant'Anna & Zhao 2020)

**Assumption.** Conditional PT, AND *either*  $\hat{\mu}_0$  or  $\hat{p}(X)$  is correct (not both required).

**Run — by hand (in wide-FD form, phat/dy\_hat already computed):**

```
gen dr_t = ever_treated * (dy - dy_hat) / 'pr_treat'  
gen dr_c = (1 - ever_treated) * (phat / (1 - phat)) * (dy - dy_hat) / 'pr_treat'  
quietly sum dr_t  
local mt = r(mean)  
quietly sum dr_c  
display "DR_ATT=" %9.0fc 'mt' - r(mean)
```

**Cross-check with drdid:**

```
drdid re $X, time(year) ivar(id) tr(ever_treated) dripw
```

**Compare to \$1,794.**

LESSON

# 6. How close did each one get?

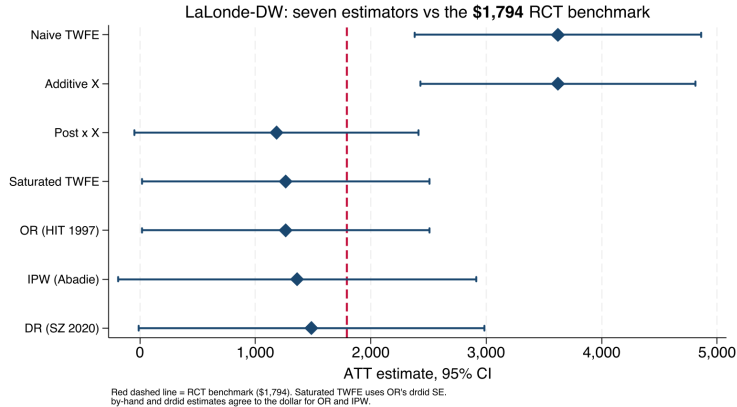
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## The comparison

| Spec                                    | ATT      | Distance from \$1,794 |
|-----------------------------------------|----------|-----------------------|
| 0. Naive TWFE (no $X$ )                 | +\$3,621 | +\$1,827              |
| A. Additive $X$                         | +\$3,621 | +\$1,827              |
| B. Post $\times X$ (## syntax)          | +\$1,184 | -\$610                |
| C. Saturated TWFE (FD)                  | +\$1,263 | -\$531                |
| OR by hand = <code>drdid reg</code>     | +\$1,263 | -\$531                |
| D. IPW by hand = <code>drdid ipw</code> | +\$1,361 | -\$433                |
| E. DR by hand                           | +\$1,496 | -\$298                |
| DR <code>drdid dripw</code>             | +\$1,485 | -\$309                |
| <b>RCT benchmark</b>                    | +\$1,794 | —                     |

*Naive = Additive- $X$ : the linear  $X$  control is absorbed. DR closest at \$298 below. By-hand = `drdid` to the dollar for OR and IPW; DR within \$11.*

# Seven specs, one benchmark



*Naive and Additive-X CIs sit entirely above \$1,794. Every design-aware spec contains it.*

## DR lands closest; the others bracket it

- **DR** (\$1,496) sits **\$298 below** — closest observational estimator.
- **IPW** (\$1,361) sits \$433 below.
- **OR / Saturated TWFE** (\$1,263) sits \$531 below — two paths, same number.
- **Post** $\times X$  (\$1,184) sits \$610 below — still misses  $\text{Bias}_{\text{HTE}}$ .
- **Additive**  $X$  (\$3,621) doesn't move off Naive (\$3,621) — the FE absorbed the linear  $X$ .

*“Doubly robust” = robust to one misspecification, not best in every sample. DR’s 298 below is selection on unobservables — the residual no observational  $X$  can close.*

## What this lab teaches you

- The **ATT is fixed** ( $\approx +\$1,794$ ). The controls only change the bias.
- **Additive  $X$  barely helps:** \$3,621 vs naive \$3,621. You need a design-aware estimator.
- $\text{Post} \times X$ , OR, IPW, DR offer four different uses of the same  $X$ , all closer to truth.
- On this dataset, **DR-DiD lands closest** at \$298 below \$1,794.
- No method is magic. Each rests on an assumption you can name.

## What additive $X$ actually assumed

The additive- $X$  spec  $Y_{it} = \alpha + \delta(D_i \cdot \text{post}_t) + X_i' \beta + \varepsilon$  was a **TWFE specification choice**. Two restrictions came along free of charge:

1. **Homogeneous treatment effect in  $X$ .** One scalar  $\delta$ . No room for  $\delta(X)$ .
2. **No time-varying  $X$  effect.** One  $\beta$ , used in both pre and post. If  $X$ 's relationship with  $Y(0)$  shifts between periods, the spec can't see it.

**Neither restriction is identification. Both are specification.**



## Two TWFE relaxations

We saw both in the synthetic simulation. Both are still TWFE — just less restrictive parameterizations.

**Post**  $\times X$ . Let  $\beta$  differ pre vs post:

$$Y = \alpha_1 + \alpha_2 T + \alpha_3 D + \delta(DT) + \beta_1 X + \beta_2 XT + \varepsilon$$

Relaxes restriction (2). Still imposes (1): one scalar  $\delta$ .

**Fully saturated.** Let  $\beta$  differ pre/post *and* let  $\delta$  depend on  $X$ :

$$Y = \alpha_1 + \alpha_2 T + \alpha_3 D + \delta(DT) + \beta_1 X + \beta_2 XT + \beta_3 XD + \beta_4 DTX + \varepsilon$$

Relaxes both. The ATT is now  $\delta + \beta_4 \mathbb{E}[X \mid D=1]$  — average the  $X$ -specific effect over the treated.

## The punchline

Additive  $X$  failed on LaLonde  
not because the data are weird,  
but because the **TWFE specification** was too restrictive.

**Fully saturated TWFE** = the regression form of HIT 1997 outcome regression.  
Same estimator. Same ATT. Different machinery.

*Spec C (FD with  $D \times X$  interactions) and OR-by-hand returned the same dollar: \$1,263.*

The ATT lives on the **treated**.

The controls only tell you  
whether your method  
can **see it**.

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